

College Mentorship Management System

A Digital Platform for Structured Faculty–Student Mentorship

Department of Computer Science and Engineering • Thapar Institute of Engineering and Technology



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Frontend Lead & UI/UX

Responsive React SPA, 3 role dashboards, design system & accessibility



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Backend Lead & API Architecture

Node.js & Express REST APIs, JWT auth, matching engine, conflict-free booking



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Database Lead & Integration

MongoDB schemas, data stores, meeting-notes logging & pilot validation

Introduction

Why a purpose-built mentorship platform is needed — and what we built instead

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Problem

- .Mentorship requests get buried under ad-hoc student emails, with no central record
- .Office hours are often unscheduled, forgotten, or cancelled without notice
- .Meeting outcomes are never logged — at-risk students are flagged too late

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Our Solution

- .A unified three-tier web app replacing ad-hoc email coordination
- .Auto-matches students to mentors by department and focus area
- .Conflict-free scheduling, persistent meeting logs, and admin analytics

Technologies Used

Layer	Technology
Frontend	React.js + Vite
Backend	Node.js + Express.js
Database	MongoDB Atlas + Mongoose
API	REST API + JWT Auth
Matching Logic	Rule-based allocation engine
Deployment	Render / Vercel (free tier)

System Objectives & Features

Four core pillars designed for students, faculty mentors, and academic administrators

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Intelligent Allocation

- Department & focus-area alignment
- First-available, transparent matching
- Admin override for manual rebalancing

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Conflict-Free Scheduling

- Faculty publish office-hour slots
- Students book in one click
- Server-side conflict prevention

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Meeting Logging & Notes

- Post-meeting summary notes
- Timestamped student audit trail
- Access restricted to mentor & admin

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Role-Tailored Dashboards

- Student: mentor, bookings, log
- Faculty: roster, schedule, notes
- Admin: metrics & reassignments

Prototype in Action

Live screenshots from the working Student Workspace — React frontend connected to the Node/Express API

Student Dashboard

At-a-glance mentor status, meeting count, and quick actions

The screenshot shows the 'Student Dashboard' of the 'UNIVERSITY PORTAL Mentorship Management System'. The user is logged in as 'jessica' (Student - Csed) and can click 'Logout'. The dashboard features a sidebar with navigation links: Dashboard, My Mentor, Request Mentor, Book Meeting, and Meeting History. The main content area displays a welcome message 'Welcome, jessica' and three summary cards: 'Assigned Mentor' (None), 'Meetings' (1), and 'Semester' (5). Below these are four action buttons: 'View Mentor' (See assigned mentor profile and availability), 'Request Mentor' (Submit a mentor allocation request), 'Book Meeting' (Schedule time with a mentor), and 'View History' (Review agendas, notes, and statuses).

Book Meeting

Conflict-free slot booking with mentor, date/time & agenda

The screenshot shows the 'Book Meeting' form in the 'UNIVERSITY PORTAL Mentorship Management System'. The user is logged in as 'jessica' (Student - Csed) and can click 'Logout'. The form includes a sidebar with navigation links: Dashboard, My Mentor, Request Mentor, Book Meeting, and Meeting History. The main content area has a title 'Book Meeting' and a form with three fields: 'Mentor' (a dropdown menu showing 'Dr. Amit Kumar - CSE'), 'Date and Time' (a date and time picker showing 'dd/mm/yyyy, --:-- --'), and 'Agenda' (a text area). A blue 'Book Meeting' button is at the bottom.

Prototype in Action

Meeting History

The screenshot displays the 'UNIVERSITY PORTAL Mentorship Management System' interface. On the left is a dark sidebar with the 'MMS Student Workspace' header and a menu containing 'Dashboard', 'My Mentor', 'Request Mentor', 'Book Meeting', and 'Meeting History' (which is highlighted). The main content area shows the user 'jessica' (Student · Csed) with a 'Logout' button. Below this, the 'STUDENT Meeting History' section features a table with the following data:

DATE	MENTOR	AGENDA	STATUS	NOTES
18 Sept 2026, 02:14	Dr. Amit Kumar	discuss regarding a research paper	Scheduled	No notes

1

Post-meeting summary notes

2

Timestamped student audit trail

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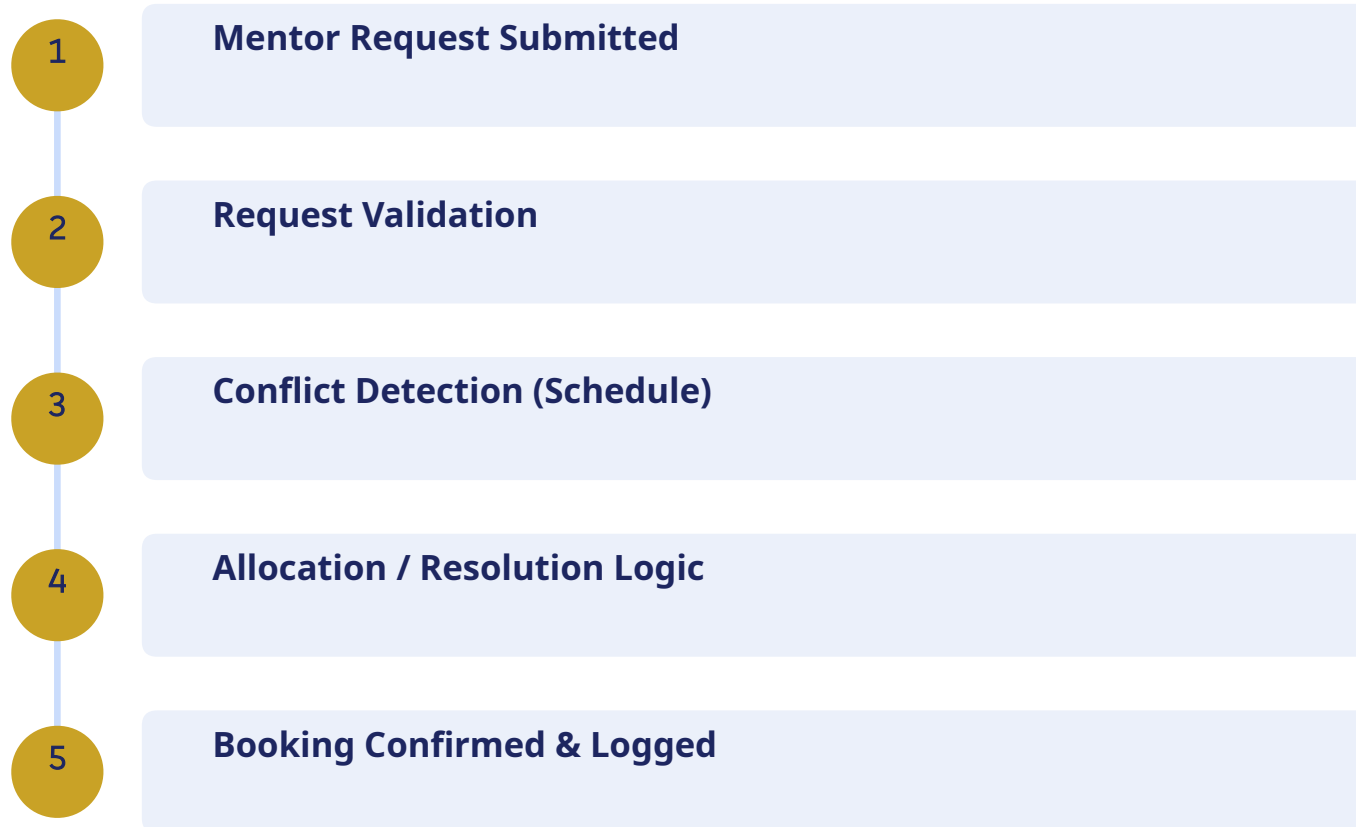
Access restricted to mentor & admin

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Status tracked — scheduled, completed, or cancelled

Methodology

Request-to-booking pipeline behind every mentorship allocation



MongoDB — Users & Allocation Store

Constraints Considered

- Department & focus-area match
- Faculty availability & office-hour slots
- Existing bookings (conflict window)
- Current mentor workload / capacity

Allocation Approach

First-available rule-based matching, ranked by department fit and current mentor load — with a manual admin .override for edge cases

MVP note: the pilot database seeds a fixed set of CSE faculty .mentors and department mappings

Week Implementation Roadmap-12

An agile, modular delivery plan ensuring on-time completion and pilot validation

WEEKS 1-2

Requirements & Schema Design

Architecture specs, ER diagrams, repo setup

WEEKS 3-5

Authentication & Allocation

JWT secure auth, student-mentor matching engine

WEEKS 6-8

Scheduling & Meeting Notes

Office-hour calendar, booking conflict engine, logs

WEEKS 9-10

Dashboards & Integration

Complete React UI for Student, Faculty & Admin

WEEK 11

Pilot Testing

Controlled pilot in CSE Dept, feedback collection

WEEK 12

Buffer & Final Report

Feedback incorporation, documentation, demo

Future Enhancements

Planned extensions beyond the MVP pilot

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Automated Notifications

.Email/push alerts for new bookings, cancellations, and mentor reassignments

2

Calendar Integration

.Two-way sync with Google Calendar / Outlook for office-hour slots

3

AI-Assisted Matching

.Weight mentor–student pairing by interests and past feedback, not just availability

4

Admin Analytics Export

.Downloadable engagement reports and CSV/PDF exports for department heads

5

Mobile App & Wider Rollout

.Native companion app and deployment across all departments campus-wide